MgO-Al_2O_3 Solid Solution: Vacancies and Density Variation

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10. Starting from a perfect MgO *crystal, a solid solution of* $\mathrm{Al_2O_3}$ *and* MgO *is prepared with an atomic proportion of* $15:85$ *of* $\mathrm{Al:Mg}$. ***Calculate: a)*** *The number of vacancies for each* Mg *atom. (Solution:* 0.088 vacancies / atoms Mg*)* ***b)*** *The* $\%$ *variation of density. (Solution:* -3%*)* ***Data:*** *Atomic masses:* $\mathrm{Al} = 26.98\ \mathrm{g/mol}$*;* $\mathrm{Mg} = 24.31\ \mathrm{g/mol}$*;* $\mathrm{O} = 16.00\ \mathrm{g/mol}$.

Solution: 0.088 vacancies / atoms Mg

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Starting matrix: Pure, perfect MgO crystal (FCC rock-salt type lattice with an $\mathrm{Mg^{2+}}$ sublattice and an $\mathrm{O^{2-}}$ sublattice).
- Atomic proportion of cations in the solid solution:

$$N_{\mathrm{Al}} : N_{\mathrm{Mg}} = 15 : 85$$

- Molar masses:
- $M_{\mathrm{Al}} = 26.98\ \mathrm{g/mol}$
- $M_{\mathrm{Mg}} = 24.31\ \mathrm{g/mol}$
- $M_{\mathrm{O}} = 16.00\ \mathrm{g/mol}$

Starting Hypotheses:

1. The $\mathrm{Al^{3+}}$ cations substitute for $\mathrm{Mg^{2+}}$ cations in the cation sublattice.
2. To maintain strict electroneutrality, every 2 $\mathrm{Al^{3+}}$ ions incorporated generate 1 magnesium cationic vacancy (V''_{Mg}) [Session 4 Slide 22].
3. The oxygen anion sublattice $\mathrm{O^{2-}}$ remains complete and free of anion vacancies.
4. The unit cell volume hardly changes because of the similarity between the radii of $\mathrm{Al^{3+}}$ (0.053 nm) and $\mathrm{Mg^{2+}}$ (0.072 nm), so the density variation is evaluated through the effective mass per lattice site.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Cationic Vacancy Balance [Session 4 Slide 22]:**

$$3\,\mathrm{Mg}^{2+} \longleftrightarrow 2\,\mathrm{Al}^{3+} + 1\,V''_{\mathrm{Mg}}$$

For each Al^{3+} atom incorporated, exactly $\frac{1}{2}$ cationic vacancy is generated:

$$N_{\text{vacancies}} = \frac{1}{2} N_{\mathrm{Al}}$$

The number of vacancies per Mg atom present is:

$$\text{Vacancies per Mg atom} = \frac{N_{\text{vac}}}{N_{\mathrm{Mg}}} = \frac{\frac{1}{2}N_{\mathrm{Al}}}{N_{\mathrm{Mg}}} = \frac{N_{\mathrm{Al}}}{2\,N_{\mathrm{Mg}}}$$

2. **Density Variation Calculation:**

Taking as a basis a set of 85 Mg atoms and 15 Al atoms:

- Number of cationic vacancies: $N_{\text{vac}} = 15/2 = 7.5$ vacancies.
- Total sites of the cation sublattice: $N_{\text{cationic sites}} = 85 + 15 + 7.5 = 107.5$ sites.
- Since in the NaCl structure the anion sublattice has exactly the same number of sites as the cation sublattice:

$$N_{\text{O atoms}} = N_{\text{cationic sites}} = 107.5\ \text{O atoms}$$

(Charge check: Positive charges $= 85 \times (+2) + 15 \times (+3) = 170 + 45 = +215$*. Negative charges* $= 107.5 \times (-2) = -215$*. Net charge* $= 0$*).*

- The mean molar mass per cation site unit in the imperfect crystal is:

$$\bar{M}_{\text{real}} = \frac{85 \cdot M_{\mathrm{Mg}} + 15 \cdot M_{\mathrm{Al}} + 107.5 \cdot M_{\mathrm{O}}}{107.5}$$

- In the perfect crystal of pure MgO, the mass per cation site is:

$$\bar{M}_{\text{ideal}} = M_{\mathrm{Mg}} + M_{\mathrm{O}} = 24.31 + 16.00 = 40.31\ \mathrm{g/mol}$$

- Percentage variation:

$$\frac{\Delta\rho}{\rho_0} = \frac{\bar{M}_{\text{real}} - \bar{M}_{\text{ideal}}}{\bar{M}_{\text{ideal}}} \times 100\%$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Number of Vacancies per Mg Atom

Taking the atomic proportion $N_{\text{Al}} = 15$ and $N_{\text{Mg}} = 85$:

$$N_{\text{vac}} = \frac{N_{\text{Al}}}{2} = \frac{15}{2} = 7.5 \text{ vacancies}$$

Ratio with respect to the number of Mg atoms:

$$\frac{N_{\text{vac}}}{N_{\text{Mg}}} = \frac{7.5}{85} = \mathbf{0.088235} \text{ vacancies / atom Mg} \approx \mathbf{0.088} \text{ vac/at Mg}$$

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2. Part b: Percentage Density Variation

- **Total mass of the mixture per** 107.5 **cation sites:**

$$m_{\text{Mg}} = 85 \times 24.31 \text{ g/mol} = 2066.35 \text{ g}$$

$$m_{\text{Al}} = 15 \times 26.98 \text{ g/mol} = 404.70 \text{ g}$$

$$m_{\text{O}} = 107.5 \times 16.00 \text{ g/mol} = 1720.00 \text{ g}$$

$$m_{\text{total, real}} = 2066.35 + 404.70 + 1720.00 = 4191.05 \text{ g}$$

- Mean mass per MgO lattice site:

$$\bar{M}_{\text{real}} = \frac{4191.05 \text{ g}}{107.5 \text{ sites}} = 38.9865 \text{ g/mol}$$

- **Mass of 107.5 units of pure perfect MgO:**

$$m_{\text{total, ideal}} = 107.5 \times (24.31 + 16.00) = 107.5 \times 40.31 = 4333.325 \text{ g}$$

$$\bar{M}_{\text{ideal}} = 40.31 \text{ g/mol}$$

- **Percentage Density Variation:**

$$\frac{\Delta\rho}{\rho_0} = \frac{\bar{M}_{\text{real}} - \bar{M}_{\text{ideal}}}{\bar{M}_{\text{ideal}}} \times 100\% = \frac{38.9865 - 40.31}{40.31} \times 100\%$$

$$\frac{\Delta\rho}{\rho_0} = \frac{-1.3235}{40.31} \times 100\% = \mathbf{-3.28\%} \approx \mathbf{-3.3\%}$$

Discrepancy with the official solution. The official key states -3%. The computed value is -3.28%; the key agrees only to one significant figure, so it is quoted as -3.3% rather than forced to -3%.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Origin of the Density Loss:** Even though the Al^{3+} cation ($M_{\mathrm{Al}} = 26.98$ g/mol) is slightly heavier than Mg^{2+} ($M_{\mathrm{Mg}} = 24.31$ g/mol), the need to neutralise the extra charge leaves 7.5 of the 107.5 cation sites (7%) as vacancies, which produces an overall mass deficit and a **net reduction of $\approx 3.3\%$ in the volumetric density** of the ceramic material.

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html